2.4 定轴转动中的功和能

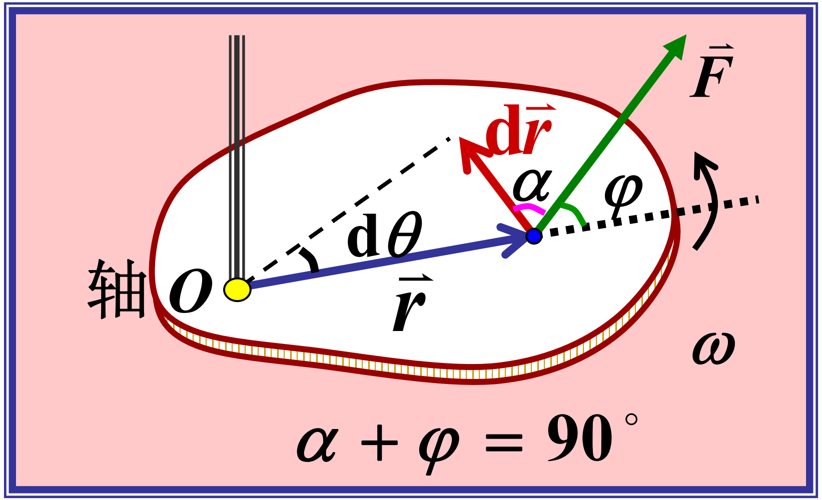
Work and Energy in Rotational Motion

2.4.1 力矩的功与功率

$$\mathrm{d}W=\vec{F}\cdot\mathrm{d}\vec{r}=F\cos\alpha\left|\mathrm{d}\vec{r}\right|=F\cos\alpha\ r\,\mathrm{d}\theta$$

$$=\underline{F\sin\varphi\, r}\,\mathrm{d}\theta=M\,\mathrm{d}\theta$$

M: 力对转轴的力矩



$$\mathrm{d}W=M\mathrm{d}\theta$$

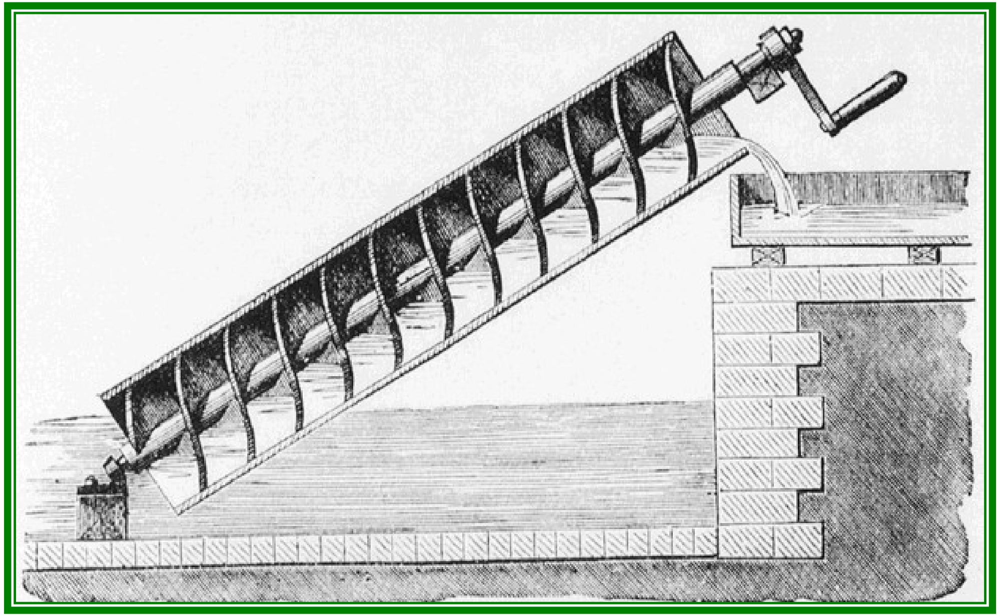
有限角位移的功： $W=\int_{\theta_1}^{\theta_2}M\mathrm{d}\theta$

有限角位移的功： $W = \int_{\theta_1}^{\theta_2} M \mathrm{d}\theta$　　力矩的空间积累

力矩的功率：

$$P = \frac{\mathrm{d}W}{\mathrm{d}t} = \frac{M\mathrm{d}\theta}{\mathrm{d}t}$$

$$P = M\omega$$



2.4.2 定轴转动动能定理

1. 转动动能 Kinetic Energy of Rotation

$$M = J\beta = J\frac{\mathrm{d}\omega}{\mathrm{d}t} = J\frac{\mathrm{d}\omega}{\mathrm{d}\theta}\frac{\mathrm{d}\theta}{\mathrm{d}t} = J\omega\frac{\mathrm{d}\omega}{\mathrm{d}\theta}$$

$$\mathrm{d}W = M\mathrm{d}\theta = J\omega\,\mathrm{d}\omega$$

$$W = \int M\mathrm{d}\theta = \int_{\omega_1}^{\omega_2} J\omega\mathrm{d}\omega = \frac{1}{2}J\omega_2^2 - \frac{1}{2}J\omega_1^2$$

定义：刚体的转动动能 $E_K = \frac{1}{2}J\omega^2$

$$\sum_i\left(\frac{1}{2}m_i v_i^2\right)=\sum_i\left(\frac{1}{2}m_i R_i^2\omega^2\right)=\frac{1}{2}\left(\sum_i m_i R_i^2\right)\omega^2$$

转动动能 $E_K=\frac{1}{2}J\omega^2$ （定轴转动刚体的转动动能）

2. 动能定理　合外力矩对刚体的功等于刚体转动动能的增量

$$W_{外}=E_{K2}-E_{K1}$$

■ 注意它与质点系动能定理的区别

$W_{外}+W_{内}=E_{K2}-E_{K1}$（一般质点系）

■ 对非定轴转动的刚体不成立

2.4.3 定轴转动的功能原理

质点系功能原理对刚体仍成立

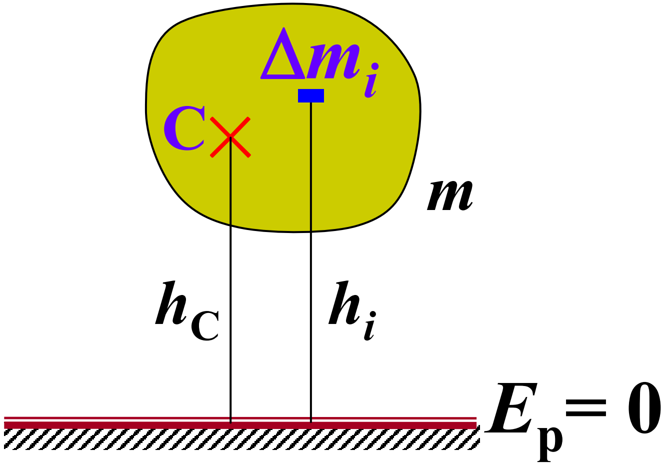
$$W_{外} + W_{内非} = (E_{k2} + E_{p2}) - (E_{k1} + E_{p1})$$

若只有保守内力做功，则机械能为常量.

刚体重力势能：

$$E_P = \sum \Delta m_i g h_i = mg \sum \frac{\Delta m_i h_i}{m}$$

$$E_P = mgh_C$$



例: 均匀直杆质量为m，长为 l，初始水平静止，轴光滑,且 $AO=l/4$ 。求：杆下摆到θ 角处时的角速度ω和轴对杆的作用力。

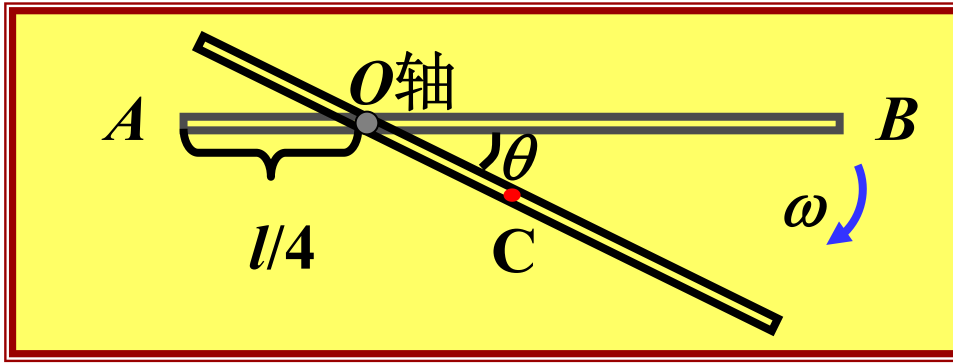
解：杆和地球组成的系统　只有重力做功，故机械能守恒

初态：$E_{k1}=0$，令 $E_{P1}=0$　末态：$E_{k2}=\dfrac{1}{2}J_O\omega^2$　$E_{p2}=-mg\dfrac{l}{4}\sin\theta$

由平行轴定理 $J_O=J_C+md^2=\dfrac{1}{12}ml^2+m(\dfrac{1}{4}l)^2=\dfrac{7}{48}ml^2$

$$\frac{1}{2}J_O\omega^2-mg\frac{l}{4}\sin\theta=0$$

$$\omega=2\sqrt{\frac{6g\sin\theta}{7l}}$$

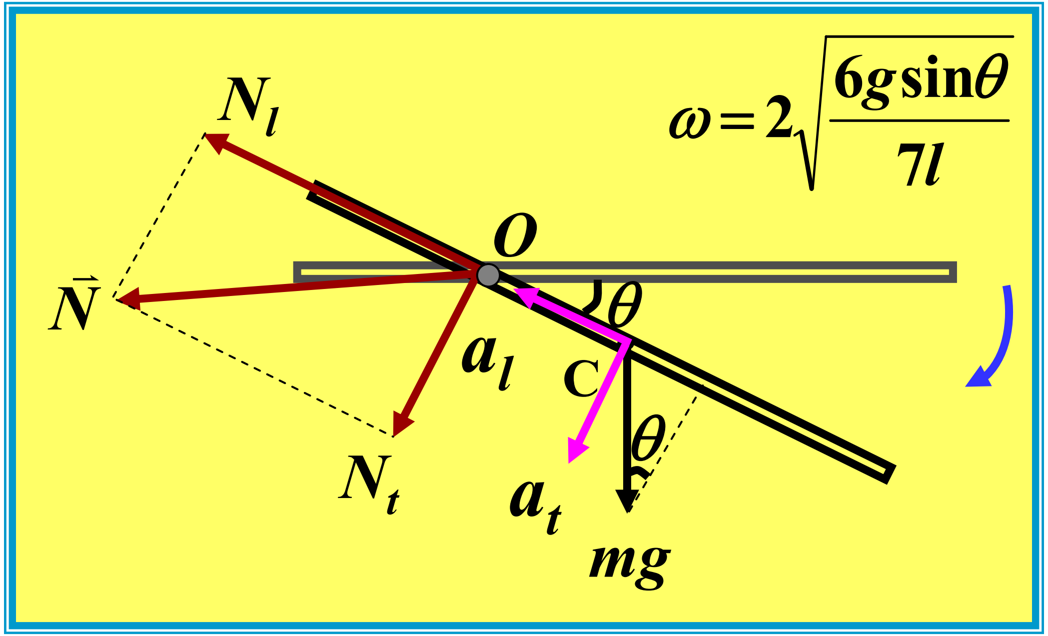


应用质心运动定理： $\vec{N} + m\vec{g} = m\vec{a}_C$

$\vec{l}$ 向 $-mg\sin\theta + N_l = ma_{Cl}$ $\vec{t}$ 向 $mg\cos\theta + N_t = ma_{Ct}$

$$a_{Cl} = \frac{l}{4}\omega^2 = \frac{6}{7}g\sin\theta$$

$$a_{Ct} = \frac{l}{4}\beta$$



$$M = J_O \beta$$

$$-mg\sin\theta + N_l = ma_{Cl}$$

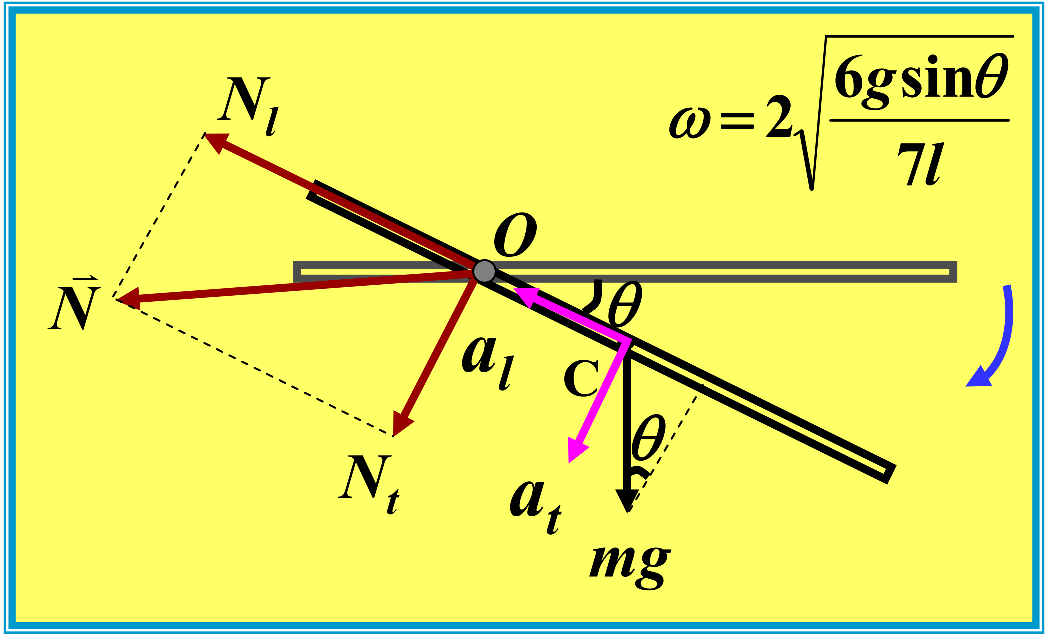
$$mg\cos\theta + N_t = ma_{Ct}$$

$$M = \frac{l}{4} mg\cos\theta$$

$$J_O = \frac{7}{48} ml^2 \qquad a_{Cl} = \frac{6}{7} g\sin\theta$$

$$\beta = \frac{12 g\cos\theta}{7l}$$

$$a_{Ct} = \frac{l}{4}\beta = \frac{3g\cos\theta}{7}$$

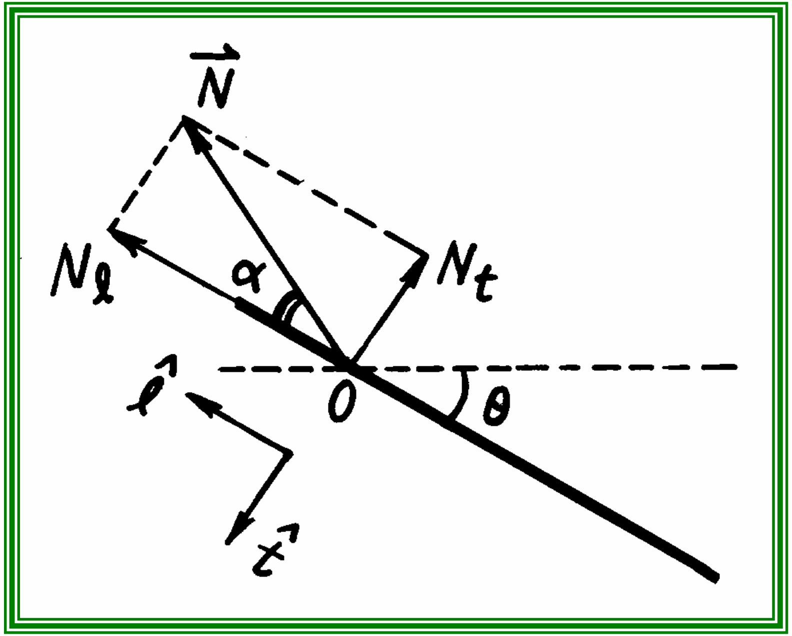


$$\begin{cases} N_l = \dfrac{13}{7} mg \sin\theta \\ N_t = -\dfrac{4}{7} mg \cos\theta \end{cases} \qquad \begin{cases} -mg\sin\theta + N_l = ma_{Cl} \\ mg\cos\theta + N_t = ma_{Ct} \end{cases} \qquad \begin{aligned} a_{Cl} &= \frac{6}{7} g \sin\theta \\ a_{Ct} &= \frac{3g\cos\theta}{7} \end{aligned}$$

$$\vec{N} = \frac{13}{7} mg \sin\theta\,\hat{l} - \frac{4}{7} mg \cos\theta\,\hat{t}$$

$$N = \frac{mg}{7}\sqrt{153\sin^2\theta + 16}$$

$$\alpha = \tan^{-1}\frac{|N_t|}{N_l} = \tan^{-1}\left(\frac{4}{13}\cot\theta\right)$$



对比质点的运动与刚体的定轴转动

质点	刚体定轴转动
$\vec{v}=\dfrac{\mathrm{d}\vec{r}}{\mathrm{d}t}$	$\omega=\dfrac{\mathrm{d}\theta}{\mathrm{d}t}$
$\vec{a}=\dfrac{\mathrm{d}\vec{v}}{\mathrm{d}t}=\dfrac{\mathrm{d}^2\vec{r}}{\mathrm{d}t^2}$	$\vec{\beta}=\dfrac{\mathrm{d}\vec{\omega}}{\mathrm{d}t}$ $\quad\beta=\dfrac{\mathrm{d}\omega}{\mathrm{d}t}=\dfrac{\mathrm{d}^2\theta}{\mathrm{d}t^2}$
m	$J=\int_V r^2\mathrm{d}m$

对比质点的运动与刚体的定轴转动

质点	刚体定轴转动
$\vec{F} = m\vec{a}$	$\vec{M} = J\vec{\beta}$
$\vec{p} = m\vec{v}$	$\vec{p} = \sum_i \Delta m_i \vec{v}_i$
$\vec{L} = \vec{r} \times \vec{p}$	$\vec{L} = J\vec{\omega}$
$\vec{F} = \frac{\mathrm{d}\vec{p}}{\mathrm{d}t} = \frac{\mathrm{d}(m\vec{v})}{\mathrm{d}t}$	$\vec{M} = \frac{\mathrm{d}\vec{L}}{\mathrm{d}t} = \frac{\mathrm{d}(J\vec{\omega})}{\mathrm{d}t}$
动量守恒 合外力为零，动量守恒	角动量守恒 合外力矩为零，角动量守恒

对比质点的运动与刚体的定轴转动

质点	刚体定轴转动
$\mathrm{d}W=\vec{F}\cdot\mathrm{d}\vec{r}=F_s\mathrm{d}s$	$\mathrm{d}W=M\mathrm{d}\theta$
$E_\mathrm{k}=\frac{1}{2}mv^2$	$E_\mathrm{k}=\frac{1}{2}J\omega^2$
$W_{外}+W_{内}=E_{\mathrm{K}2}-E_{\mathrm{K}1}$	$W_{外}=E_{\mathrm{K}2}-E_{\mathrm{K}1}$